Preliminary Report
Hurricane Felicia
14 - 22 July 1997

Miles B.Lawrence
National Hurricane Center
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a. Synoptic History

The first indication of Hurricane Felicia was the detection of a large area of increasing thunderstorms centered several hundred miles south of Manzanillo, Mexico on July 13th. On the 14th, this disturbed weather became better organized with a banding-type cloud pattern and developed into Tropical Depression Eight-E. Fig. 1 shows a plot of the best track of Felicia and this track is tabulated in Table 1. Felicia's movement during its existence was toward the west-northwest at forward speeds varying from 5 to 15 knots.

For the first day or so, there was little additional development as the depression experienced northeasterly wind shear, partially associated with the outflow from Hurricane Enrique. But on the 15th, the system was upgraded to Tropical Storm Felicia as Enrique's outflow retreated. A burst of deep convection on the 16th was evidence of further development and Felicia became a hurricane on the 17th when there was a suggestion of the formation of an eye between two interlocking convective bands. Felicia leveled off as a 65-knot hurricane for 24 hours under northwesterly shear caused by a nearby upper-level trough.

On the 18th, an eye became better defined and a period of intensification began that culminated with estimated 115-knot sustained wind speeds on the 19th. The hurricane's maximum wind speed remained 115 knots for most of the 19th and then began a continuous decrease as a result of colder sea surface temperatures and a hostile synoptic environment caused by another approaching upper-level trough.

Felicia moved west of 140 degrees west longitude on the 21st and the responsibility for issuing advisories was transferred to the National Weather Service's Central Pacific Hurricane Center at Honolulu at that time. Felicia was reduced to a swirl of low clouds on the 22nd in the central Pacific Ocean.

b. Meteorological Statistics

Figures 2 and 3 show curves of minimum sea-level pressure and maximum one-minute surface wind speed, respectively, as a function of time. Satellite data plotted in these figures are based on the Dvorak satellite intensity estimating technique as applied at the Tropical Analysis and Forecast Branch (TAFB), the Satellite Analysis Branch (SAB) and the U.S. Air Force Global

Weather Center (AFGWC). As is the case for most eastern Pacific tropical cyclones, all of the tracking and intensity estimates of Felicia were based on satellite data.

c. Casualty and Damage Statistics

Felicia did not affect land and there were no known casualties or damage.

d. Forecast and Warning Critique

Average official track forecast errors were 8, 33, 66, 95, 125, and 210 nautical miles, respectively, for the 0-, 12-, 24-, 36-, 48-, and 72-hour forecast periods. These errors are slightly smaller than the 1988-1995 average errors except at 72 hours where the errors are slightly larger. The wind speed forecast errors were rather large for the two days preceding the strengthening to 115 knots. The 48-hour forecast error for the forecast issued at 0300 GMT on the 18th was -55 knots, as there was a failure to recognize the potential for the significant strengthening that subsequently occurred.

Table 1. Best track, Hurricane Felicia, 14 - 22 July 1997.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed(kt)	Stage
14/1200	9.7	109.0	1006	30	tropical depression
1800	10.4	110.1	1006	30	"
15/0000	10.9	111.1	1006	30	"
0600	11.3	112.0	1005	30	"
1200	11.7	112.5	1005	30	"
1800	12.1	113.0	1004	35	tropical storm
16/0000	12.4	113.7	1003	35	"
0600	12.5	114.5	1000	40	"
1200	12.5	115.1	996	50	"
1800	12.6	115.5	990	55	"
17/0000	12.9	116.0	988	60	"
0600	13.4	116.9	987	65	hurricane
1200	13.9	117.9	987	65	"
1800	14.4	119.0	987	65	"
18/0000	14.8	120.1	985	65	"
0600	15.1	121.3	979	75	"
1200	15.3	122.5	968	95	"
1800	15.4	123.9	960	100	"
19/0000	15.5	125.3	956	110	"
0600	15.6	126.7	951	115	"
1200	15.7	128.1	948	115	"
1800	15.8	129.5	951	115	"
20/0000	16.0	131.0	954	110	"
0600	16.3	132.5	957	105	"
1200	16.6	134.1	960	100	"
1800	16.9	135.6	965	95	"
21/0000	17.1	137.2	972	90	"
0600	17.3	138.9	980	70	"
1200	17.4	140.8	990	55	tropical storm
1800	17.5	143.0	995	45	"
22/0000	17.6	145.4	1000	40	"
0600	17.6	147.9	1005	30	tropical depression
1200	17.7	150.2	1007	25	"
19/1200	15.7	128.1	948	115	minimum pressure

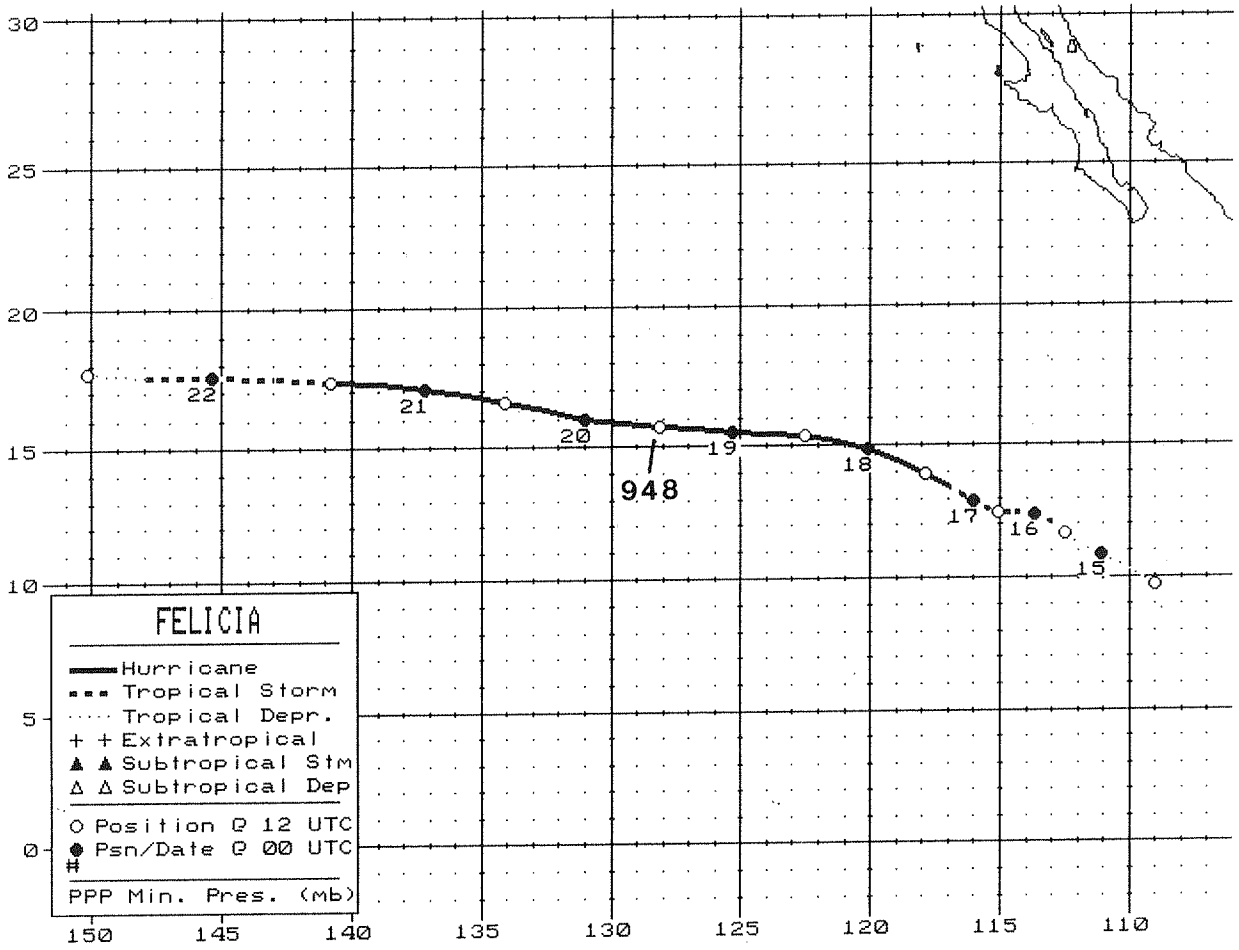

Fig. 1. Best track positions for Hurricane Felicia, 14-22 July 1997.

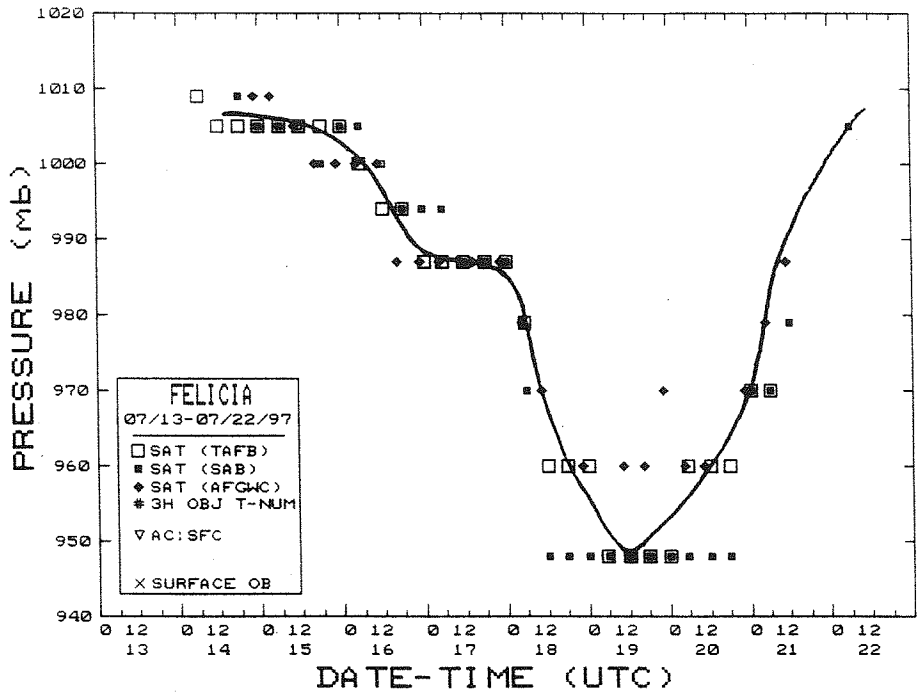

Fig. 2. Best track minimum sea level pressure curve for Hurricane Felicia.

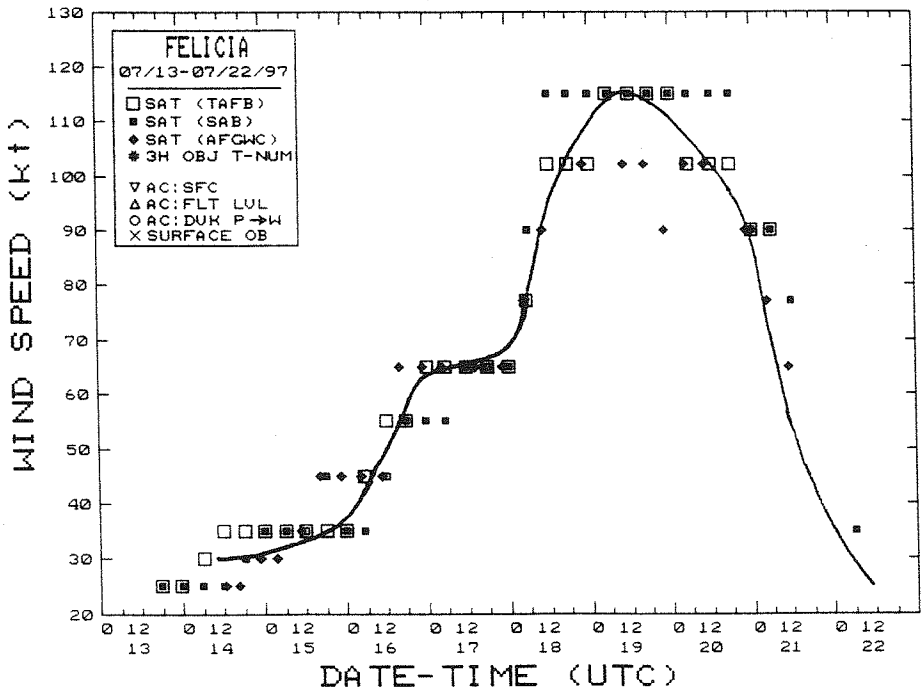

Fig. 3. Best track maximum one-minute wind speed curve for Hurricane Felicia.